Introduction to Probability and Statistics
(EE 354 / CE 361)

Discrete RVs: Variance and SD for RVs, Joint PMFs

Expectation/Mean of a Function of a Random Variable

- We can extend the concepts of PMF and expectation developed so far for a single RV to multiple RVs

In particular, to obtain $p_Y(y)$ for any y, we add the probabilities of all values of x such that $g(x) = y$:

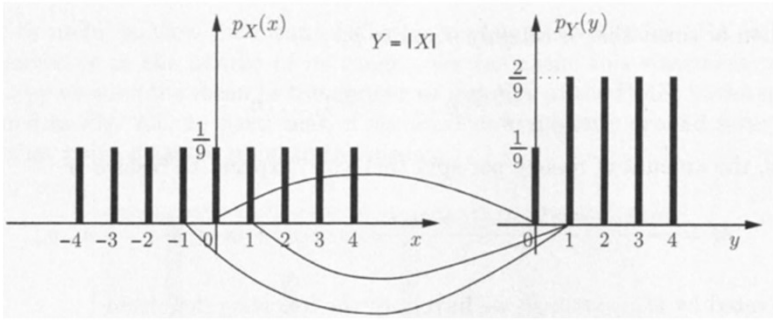
$$p_Y(y) = \sum_{\{x \mid g(x)=y\}} p_X(x).$$

- Now, the expectation of Y can be obtained by applying the definition to p_Y

$$\mathbf{E}[Y] = \sum_y y\, p_Y(y)$$

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$

$$p_Y(y) = \begin{cases} 2/9. & \text{if } y = 1, 2, 3, 4, \\ 1/9. & \text{if } y = 0. \\ 0. & \text{otherwise.} \end{cases}$$



Expectation/Mean of a Function of a Random Variable

If $Y = g(X)$ is a function of random variable X,

- **The Harder Approach for Calculating Mean (Last slide):**
  - Calculate the PMF of Y using the PMF of X
  - Now, the expectation of Y can be obtained by applying the definition to p_Y: $\mathbf{E}[Y] = \sum_y y\, p_Y(y)$
- **An Easier Approach for Calculating Mean:**

Expected Value Rule for Functions of Random Variables

Let X be a random variable with PMF p_X, and let $g(X)$ be a function of X. Then, the expected value of the random variable $g(X)$ is given by

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x).$$

  - Allows us to calculate the expectation of Y directly from the PMF of X

Mean of a Linear Function of a Random Variable

Let X be a random variable and let

$$Y = aX + b.$$

where a and b are given scalars. Then,

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b,$$

$$\mathbf{E}[Y] = \sum_x (ax + b)\, p_X(x)$$

$$= a \sum_x x\, p_X(x) + b \sum_x p_X(x)$$

$$= a\, \mathbf{E}[Y] + b$$

Caution: unless g(X) is a linear function, it is not generally true that $\mathbf{E}[g(X)]$ is equal to $g(\mathbf{E}[X])$. In general, $\mathbf{E}[g(X)] \neq g(\mathbf{E}[X])$

Today's Agenda

- Variance
  - Motivation, definition and intuition
  - Bernoulli RV
  - Standard Deviation

- (time permitting) Introduction to Joint PMFs

Variance of a Random Variable

- Definition:

Variance

The variance $\text{var}(X)$ of a random variable X is defined by

$$\text{var}(X) = \mathbf{E}\left[\left(X - \mathbf{E}[X]\right)^2\right],$$

and can be calculated as

$$\text{var}(X) = \sum_x \left(x - \mathbf{E}[X]\right)^2 p_X(x).$$

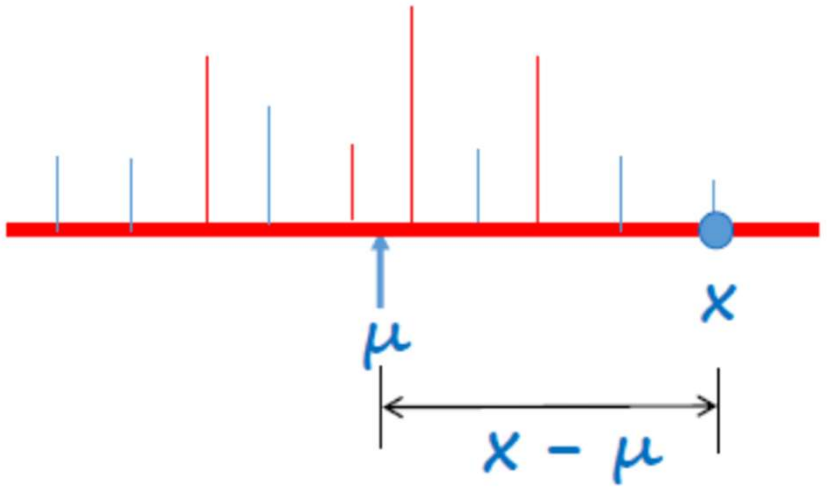
It is always nonnegative. Its square root is denoted by σ_X and is called the **standard deviation**.

- Interpretation:
  - A measure of the spread/dispersion of a PMF around its mean

Variance of a Random Variable

- Interpretation:
  - A measure of the spread/dispersion of a PMF around its mean

Consider 2 RVs X and Y with the same mean μ



$$\text{var}(X) > \text{var}(Y)$$

Variance of a Random Variable

- Definition:

Variance

The variance $\mathrm{var}(X)$ of a random variable X is defined by

$$\mathrm{var}(X) = \mathbf{E}\left[\left(X - \mathbf{E}[X]\right)^2\right],$$

- Calculation
  - Based on the Expected Value Rule for Functions of a Random Variable, the variance of a random variable X can be calculated as:

$$\mathrm{var}(X) = \sum_x \left(x - \mathbf{E}[X]\right)^2 p_X(x).$$

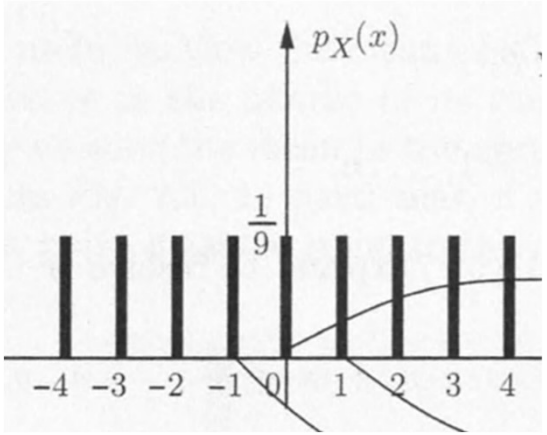
Variance of a Random Variable

- Calculation
  - Based on the Expected Value Rule for Functions of a Random Variable, the variance of a random variable X can be calculated as:

$$\mathrm{var}(X) = \sum_x \big(x - \mathbf{E}[X]\big)^2 p_X(x).$$

- Example:

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$



Variance of a Random Variable

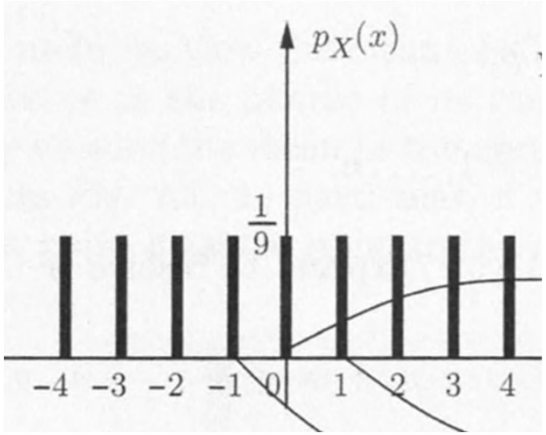
- Calculation
  - Based on the Expected Value Rule for Functions of a Random Variable, the variance of a random variable X can be calculated as:

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- Example:

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$

$$\begin{aligned} \text{var}(X) &= \mathbf{E}\left[\big(X - \mathbf{E}[X]\big)^2\right] \\ &= \sum_x \big(x - \mathbf{E}[X]\big)^2 p_X(x) \\ &= \frac{1}{9} \sum_{x=-4}^{4} x^2 \qquad (\text{since } \mathbf{E}[X] = 0) \\ &= \frac{1}{9}(16 + 9 + 4 + 1 + 0 + 1 + 4 + 9 + 16) \\ &= \frac{60}{9}. \end{aligned}$$



Standard Deviation of a Random Variable

- Definition

The variance $\text{var}(X)$ of a random variable X is defined by

$$\text{var}(X) = \mathbf{E}\left[\left(X - \mathbf{E}[X]\right)^2\right],$$

It is always nonnegative. Its square root is denoted by σ_X and is called the **standard deviation**.

- Interpretation
  - A measure of the spread/dispersion of a PMF around its mean
  - In some cases, standard deviation is easier to interpret because it has the same units as X.

Properties of Variance

$$\text{var}(X) \geq 0$$

- Since $(X - \mathbf{E}[X])^2$ can only take on nonnegative values, the variance is always nonnegative

Let X be a random variable and let

$$Y = aX + b.$$

where a and b are given scalars. Then.

$$\text{var}(Y) = a^2 \, \text{var}(X).$$

- How? Can we prove the second property?

Properties of Variance

Let X be a random variable and let

$$Y = aX + b.$$

where a and b are given scalars. Then.

$$\text{var}(Y) = a^2 \, \text{var}(X).$$

$$\begin{aligned}
\text{var}(Y) &= \sum_x \big(ax + b - \mathbf{E}[aX + b]\big)^2 p_X(x) \\
&= \sum_x \big(ax + b - a\mathbf{E}[X] - b\big)^2 p_X(x) \\
&= a^2 \sum_x \big(x - \mathbf{E}[X]\big)^2 p_X(x) \\
&= a^2 \, \text{var}(X).
\end{aligned}$$

Properties of Variance

$$\mathrm{var}(X) = \mathbf{E}[X^2] - \big(\mathbf{E}[X]\big)^2.$$

$$\begin{aligned}
\mathrm{var}(X) &= \sum_x \big(x - \mathbf{E}[X]\big)^2 p_X(x) \\
&= \sum_x \Big(x^2 - 2x\mathbf{E}[X] + \big(\mathbf{E}[X]\big)^2\Big) p_X(x) \\
&= \sum_x x^2 p_X(x) - 2\mathbf{E}[X] \sum_x x p_X(x) + \big(\mathbf{E}[X]\big)^2 \sum_x p_X(x) \\
&= \mathbf{E}[X^2] - 2\big(\mathbf{E}[X]\big)^2 + \big(\mathbf{E}[X]\big)^2 \\
&= \mathbf{E}[X^2] - \big(\mathbf{E}[X]\big)^2.
\end{aligned}$$

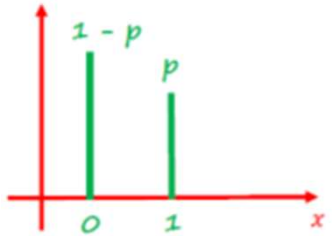
Variance of a Bernoulli Random Variable

- Experiment
  - The toss of a coin which comes up heads with probability p and tails with probability $(1 - p)$.
- Sample Space
  - {Head, Tail}

$$X = \begin{cases} 1. & \text{if a head,} \\ 0. & \text{if a tail.} \end{cases}$$

- Random Variable
  - The RV takes value 1 if the outcome is head and takes the value 0 if the outcome is a tail.
- PMF

$$p_X(k) = \begin{cases} p, & \text{if } k = 1, \\ 1 - p, & \text{if } k = 0. \end{cases}$$



- Mean

$$\mathbf{E}[X] = 1(p) + 0(1 - p) = p$$

- Variance

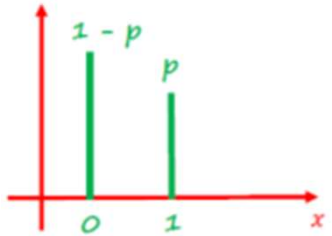
Variance of a Bernoulli Random Variable

- Experiment
  - The toss of a coin which comes up heads with probability p and tails with probability $(1 - p)$.
- Sample Space
  - {Head, Tail}
- Random Variable
  - The RV takes value 1 if the outcome is head and takes the value 0 if the outcome is a tail.

$$X = \begin{cases} 1. & \text{if a head,} \\ 0. & \text{if a tail.} \end{cases}$$

- PMF

$$p_X(k) = \begin{cases} p, & \text{if } k = 1, \\ 1 - p, & \text{if } k = 0. \end{cases}$$



- Mean

$$\mathbf{E}[X] = 1(p) + 0(1 - p) = p$$

- Variance

$$\mathbf{E}[X^2] = 1^2 \cdot p + 0 \cdot (1 - p) = p,$$

$$\boxed{\text{var}(X) = \mathbf{E}[X^2] - \big(\mathbf{E}[X]\big)^2 = p - p^2 = p(1 - p).}$$

Example: Variance

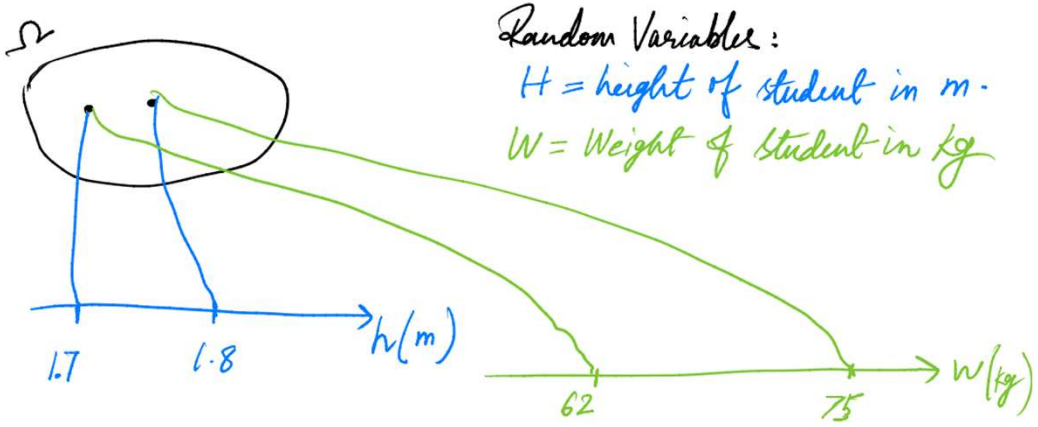
- Random Variable
  - Minimum of two independent rolls of a fair tetrahedral die

- Variance?

Recall: Multiple Random Variables on the same Sample Space

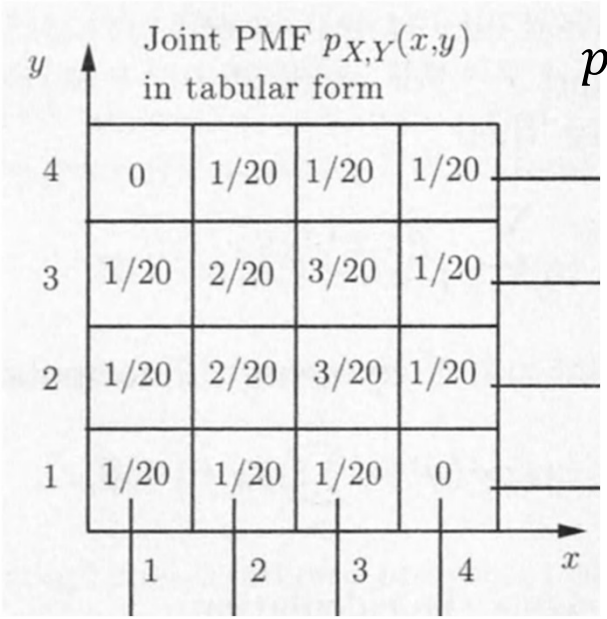
- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class



Multiple Random Variables and Joint PMF

- We can extend the concepts of PMF and expectation developed so far for a single RV to multiple RVs
- Consider two discrete random variables X and Y associated with the same experiment.
  - The probabilities of the values that X and Y can take are captured by the joint PMF of X and Y, denoted by $p_{X,Y}$
  - In particular, if (x, y) is a pair of possible values of X and Y, the probability mass of (x, y) is the probability of the event $\{X = x, Y = y\}$



$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y)$$
$$= \mathbf{P}(\{X = x\} \cap \{Y = y\})$$
$$= \mathbf{P}(X = x \text{ and } Y = y)$$

